# Keelan O'Doherty

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## EDUCATION

**University of Michigan** Ann Arbor, MI · Aug 2026 – May 2028

*Master of Science in Engineering (MSE), Aerospace Engineering, Space Systems*

- Relevant Coursework: Spacecraft Technology, Space Policy & Management, Linear Systems Theory, Management of Space Systems Design, Astrodynamics

**University of Alabama, Honors College** Tuscaloosa, AL · Aug 2022 – May 2026

*Bachelor of Science (BS) in Aerospace Engineering, Minor in Mathematics*

- **Honors GPA 3.64/4.0** · Dean's List: every semester, Fall 2022 – Spring 2025
- Scholarships: Full-Ride Academic, Presidential, Engineering, Taylor Lewan Offensive Line Scholarship

**Florence University of the Arts, Study Abroad** Florence, Italy · May 2024

## ENGINEERING EXPERIENCE

**Northrop Grumman Sponsored Lunar Cargo Module (CAT-LIT)** Ann Arbor, MI · Sep 2026 – Present

*Systems Engineer, SPACE 582 Spacecraft Design (Northrop Grumman sponsored)*

- Drafted 48 Level 0 and Level 1 requirements for a standardized lunar cargo module (8,000 kg cargo, 2,000 kg dry, 5/7/8 m variants) compatible with Starship and Blue Moon, each traced to a parent with a verification method and success criteria (NASA SP-2016-6105)
- Set the mission statement and six key performance parameters with threshold and objective values for the December mission design review and system requirements review
- Authored the cargo history and interface standards section of the lunar cargo delivery survey; checked lander and launch vehicle envelope and load data against the Falcon, Vulcan, Ariane 6, New Glenn and Starship user guides

**University of Alabama Aerodynamics Research Laboratory (AAERO)** Tuscaloosa, AL · Dec 2024 – May 2026

*Undergraduate Research Assistant, Vertical Flight Society Chapter*

- Designed, SLA 3D-printed, and bench-tested 4 UAV propeller geometries at 10,000 RPM to characterize thrust against acoustic performance
- Built the test setup and authored run procedures using load-cell thrust instrumentation and microphone acoustic profiling, structured for repeatable comparison across geometries
- Correlated measured results against Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) predictions to identify model-to-hardware divergence; presented findings at the URCA Conference 2025

**Fixed-Wing eVTOL Aircraft - Design, Build & Flight Test** Senior Capstone · Aug 2025 – May 2026

- Modeled the full ORCA airframe in SolidWorks from scratch for 3D printing, adapted from an existing VTOL design (fuselage, wings, spars, control surfaces, motor mounts), including a modular payload nose for optical, IR, and thermal sensor swaps
- Served as team pilot and ran the flight test campaign: phased hover and vertical-to-forward transition trials validating stability, control response, and endurance
- Modeled propulsion tilt-path dynamics and servo actuation sequencing in MATLAB/Simulink; validated the power budget against measured draw

**Wilson & Company** Phoenix, AZ · Jul 2024 – Aug 2024

*Engineering Intern*

- Gained hands-on experience with construction materials and techniques through 50+ hours of on-site exposure, documenting nonconformances and as-built conditions

## ADDITIONAL EXPERIENCE

**Watercraft Restoration - Owner & Technician** Phoenix, AZ · Feb 2021 – Aug 2026

- Restored 65+ watercraft end to end, including 30+ full engine rebuilds and 30+ complete electrical rewires, from teardown through functional test
- Fabricated wire harnesses and mechanical subassemblies to torque specification; wrote repeatable teardown, build, and checkout procedures that reduced rework on later units

**Pi Kappa Alpha Fraternity - Class President; Sports Chair** Tuscaloosa, AL · 2022 – 2026

## TECHNICAL PROJECTS

- **Embedded Vehicle Control System** - built Arduino-based control with GSM/IoT telemetry (SIM7000A) over LTE CAT-M1 for remote monitoring and actuation; designed power distribution using transistor logic and relay control, and wired the physical harness

## TECHNICAL SKILLS

**CAD:** SolidWorks (90-part ORCA airframe assembly with wing, tail and fuselage subassemblies built for 3D printing; motor-mount and spar revisions; nozzle geometry exported to STEP for CFD), FDM 3D printing, wire harness design

**Mission & Orbit Analysis:** STK 12 (sun-synchronous and mid-inclination coverage scenarios with rectangular and conic sensors, revisit and N-asset figures of merit; Astrogator lunar transfer with differential-corrector TLI and B-plane targeted LOI; low-thrust spiral with a custom propagator), patched-conic and Hohmann analysis in MATLAB

**CFD & Simulation:** ANSYS Fluent (axisymmetric density-based nozzle flow, SST k-omega, four-level mesh refinement to 137k cells; NACA 2412 with Spalart-Allmaras and SST; heated-channel convection), MATLAB/Simulink (DC-8 state-space pitch and roll-yaw control loops, quasi-1D nozzle solver, intercept simulation, solid and liquid stage sizing)

**Programming & Tools:** Python (NumPy, SciPy, Jupyter: laminate and buckling analysis, design searches), MATLAB, Arduino/C++, Git/GitHub, Excel, Microsoft 365

**Additional:** Languages: Spanish · Musical: Piano, Cello, Guitar